Lecture 7: Liquid Biopsy for Cancer Metastasis Detections

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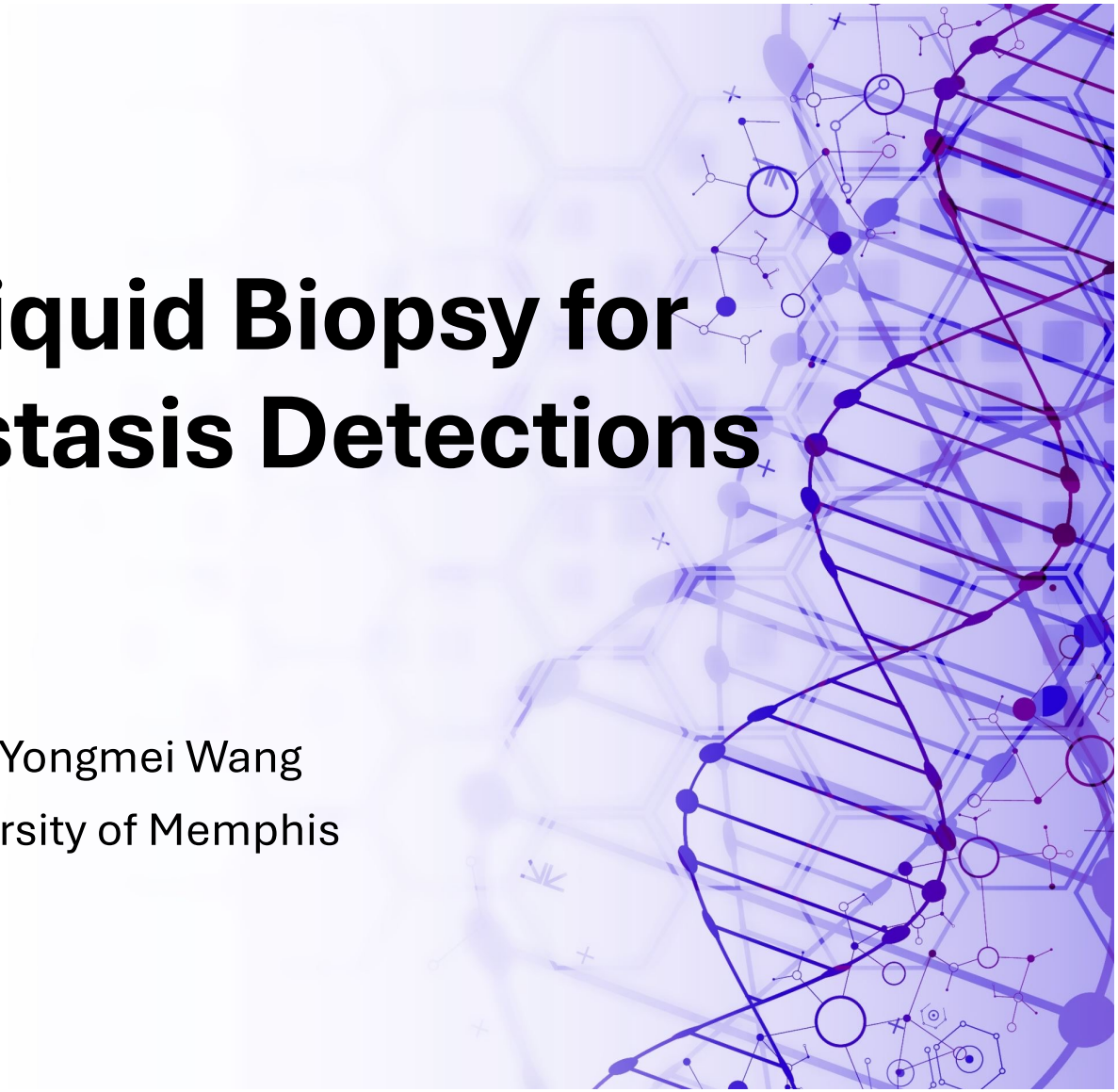
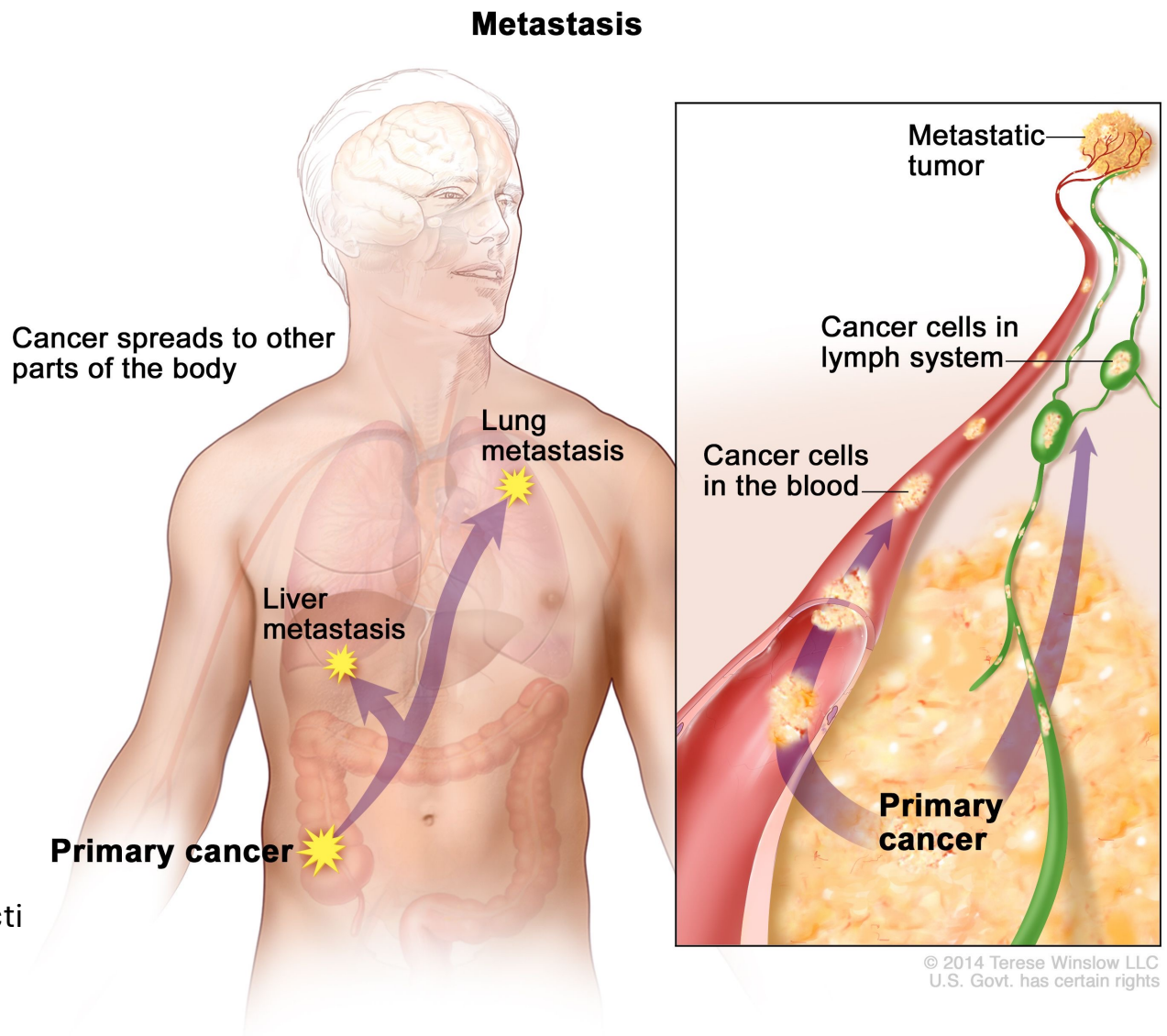


What is cancer metastasis?

- Cancer cells spread from the **primary tumor site** to form **new tumors in distant organs or tissues**.
- It is **not random**: cancer cells often spread to specific organs (e.g., breast cancer commonly spreads to **bone, brain, lung, liver**)---called organotropism
- Cancer cells after metastasis are **still the same type** (e.g., breast cancer cells in the brain are not brain cancer).
- Metastasis is responsible for the **vast majority of deaths (>90%).**

- Watch this video
- https://www.cancer.gov/publications/dictionaries/cancer-terms/def/metastasis



https://www.cancer.gov/publications/dictionaries/cancer-terms/def/metastasis

Key steps in metastasis

- **Local Invasion**
  - EMT (Epithelial-Mesenchymal Transition)
  - Breakdown of basement membrane
- **Intravasation**
  - Entry into blood or lymph vessels
  - Role of **tumor-associated macrophages (TAMs)**
- **Circulation (CTCs)**
  - Circulating Tumor Cells face immune attack & shear stress
  - Often travel in clusters
- **Extravasation**
  - Tumor cells adhere to endothelial walls and exit vessels
- **Colonization**
  - Most fail; **micrometastases** may remain dormant
  - Requires new angiogenesis, immune evasion

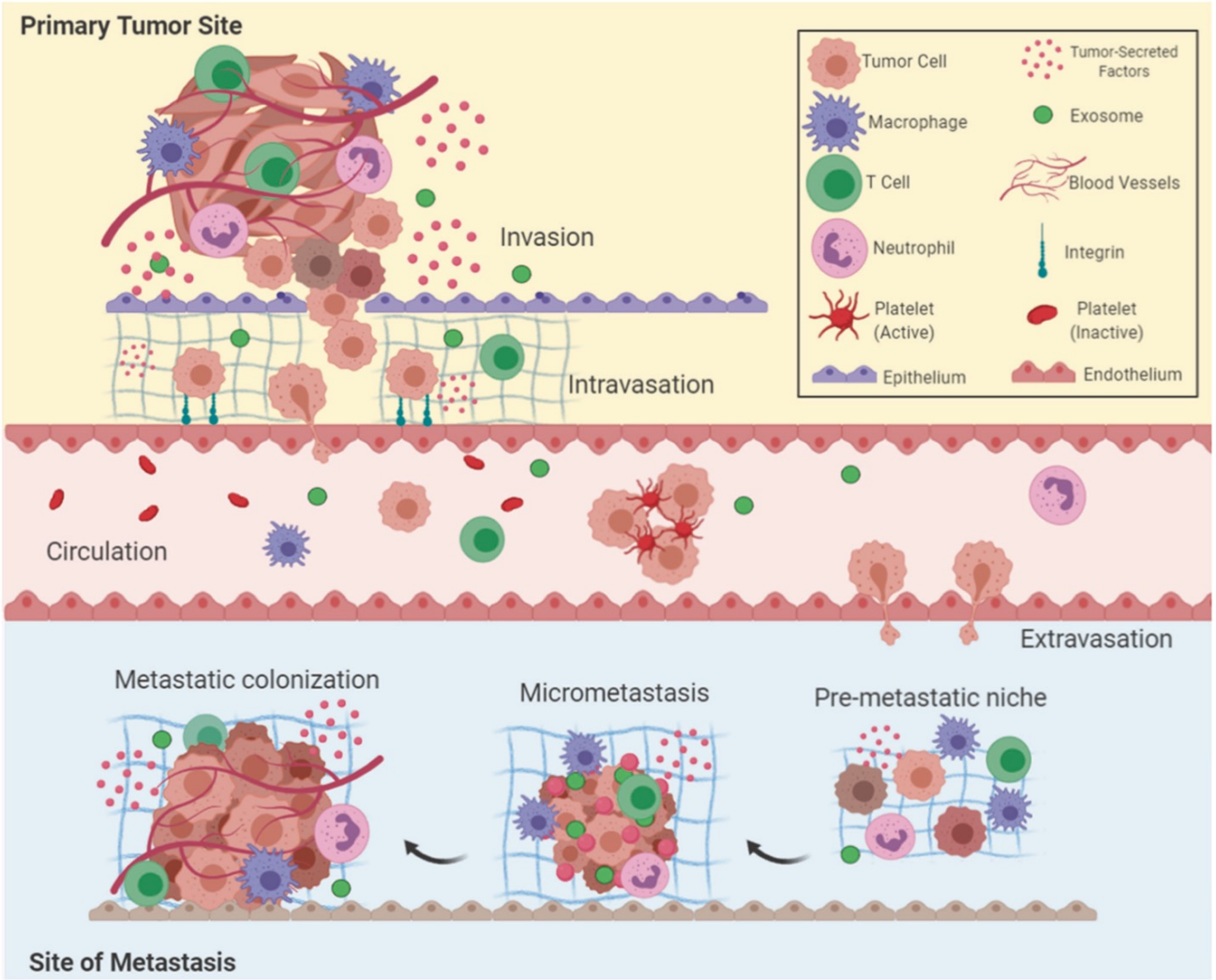
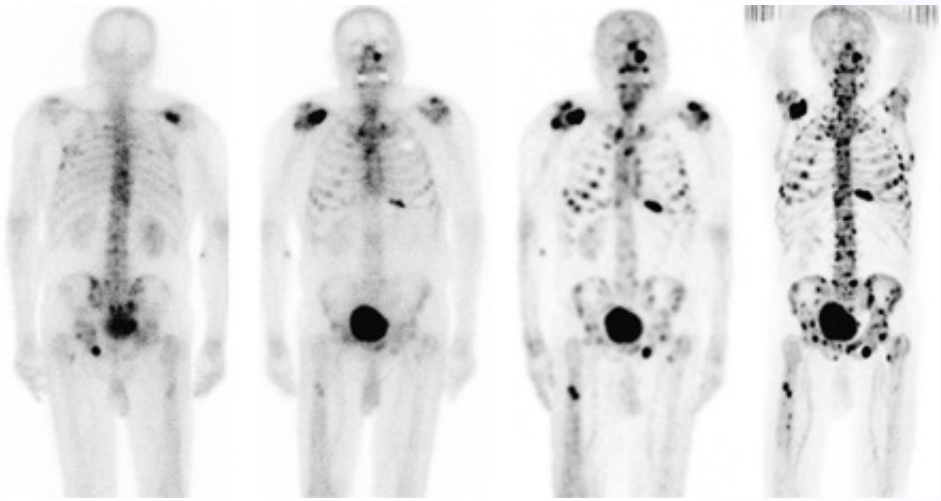


Fig. 1 Overview of the metastatic cascade: The five key steps of metastasis include invasion, intravasation, circulation, extravasation, and colonization

https://www.nature.com/articles/s41392-020-0134-x

How to detect metastasis

- Usually through image such as CT scan, MRI etc



Examples of prostate cancer patient with bone metastasis
https://jnm.snmjournals.org/content/47/2/287/F1

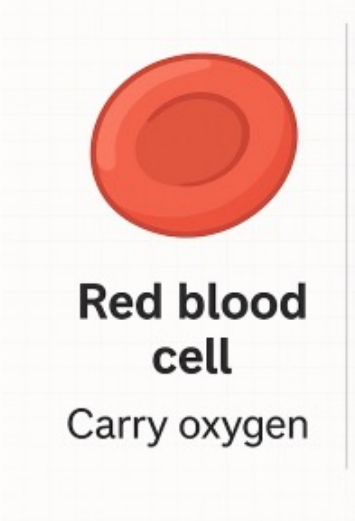
Detect Circulating Tumor Cells (CTCs)

- Idea of the existence of CTCs was proposed long time ago
- **1990s–2000s**: CTC detection re-emerged, especially with the development of enrichment and detection platforms using nanotechnology
- In 2004, the **CellSearch® system** (Veridex) became the **first FDA-cleared test** for enumerating CTCs in metastatic breast, prostate, and colorectal cancers.

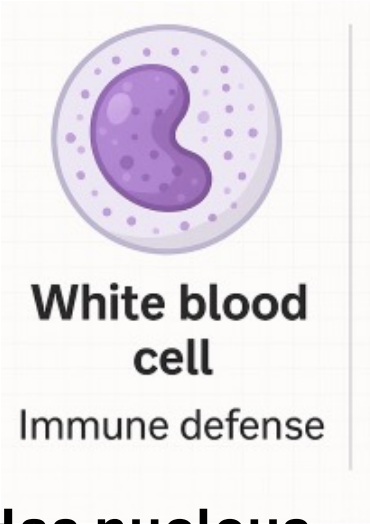
What is the blood?

- **Red blood cells (RBCs)** are by far the most numerous (~4.5–6 million per mL).
- **White blood cells (WBCs)** are far fewer (~4,000–11,000 per mL).
- **Platelets** are intermediate (~150,000–450,000 per mL).
- **CTCs** (if present in cancer patients) occur at **<10 cells per mL of blood** — extremely rare.

Differences between these types

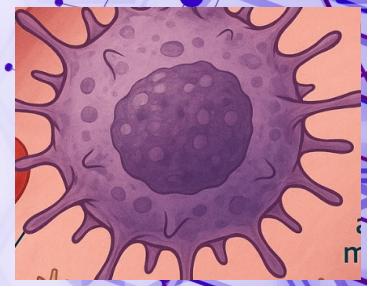


No nucleus
Size 6–8 μm
Thickness 2μm



Has nucleus
Diverse types
Size larger than RBC
8μm above

No nucleus
Cell fragments



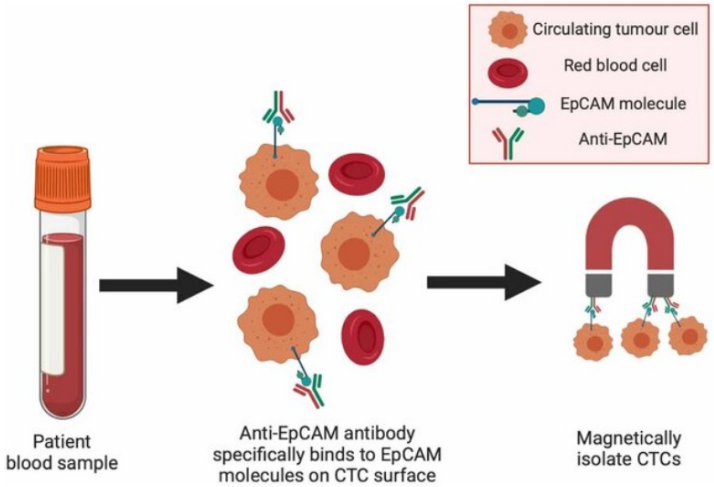
Circulating Tumor Cells (CTCs)

has nucleus
Cell origins often epithelial
Size larger than RBC

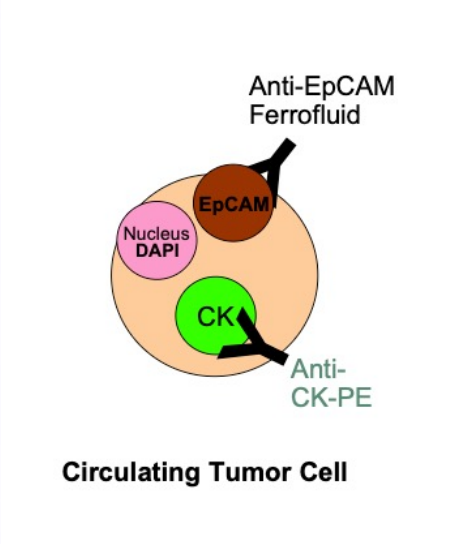
CellSearch@ https://www.cellsearchctc.com/

- In 2004 The **CellSearch® system** (Veridex) became the **first FDA-cleared test** for enumerating CTCs in metastatic breast, prostate, and colorectal cancers.
- It is the first and only clinically validated, FDA-cleared blood test for enumerating circulating tumor cells (CTCs).
- As of **late 2016**, when **Menarini Silicon Biosystems** acquired the CellSearch CTC business from Janssen Diagnostics (previously Veridex), it was reported that **approximately 300 laboratories worldwide were actively using the CellSearch CTC system**

How CellSearch works



Immunomagnetic Separation Using anti-EpCAM ferrofluids



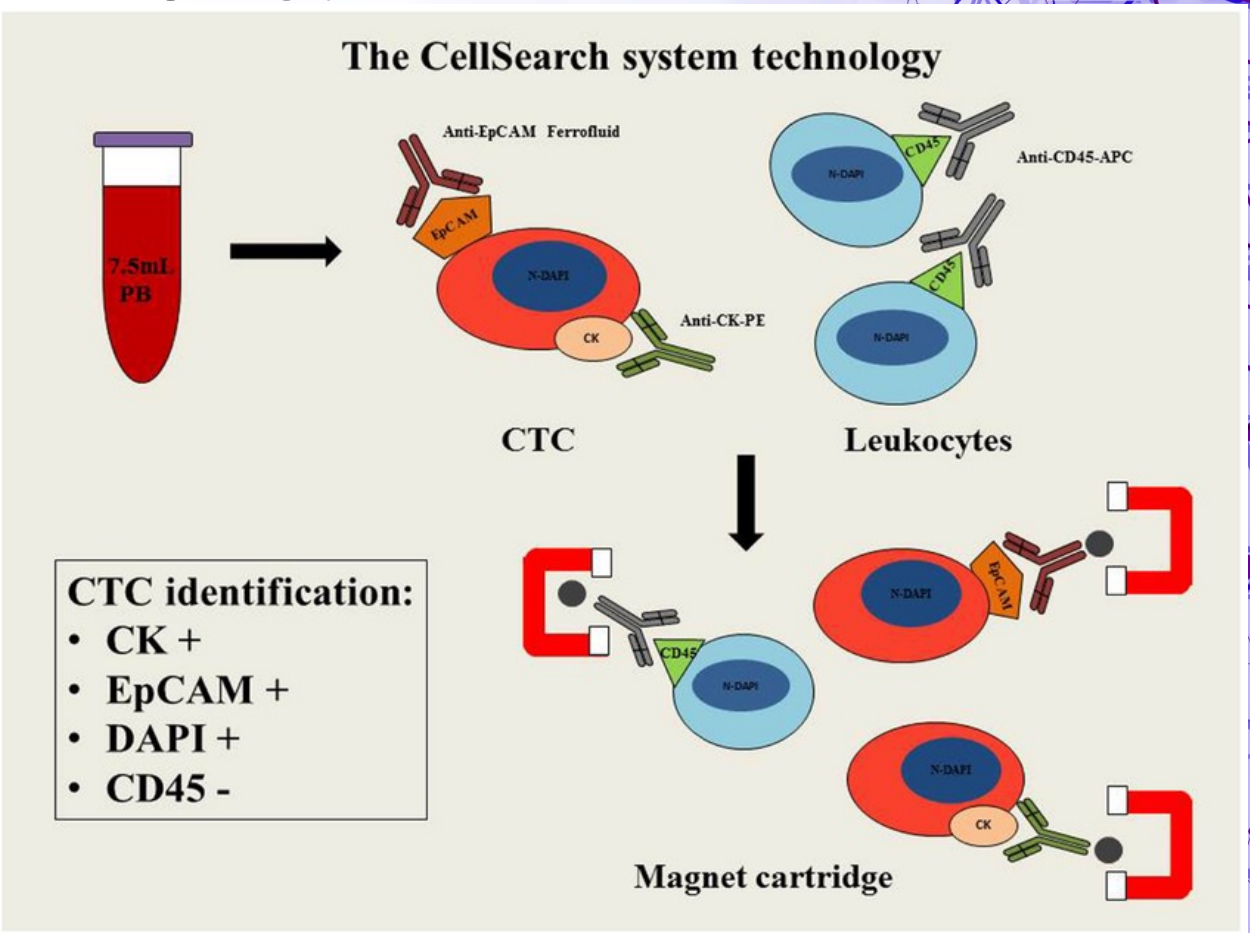
Fluorescence Staining and Identifications

- DAPI for Nucleus
- Cytoskeleton (CK+)
- CD45-

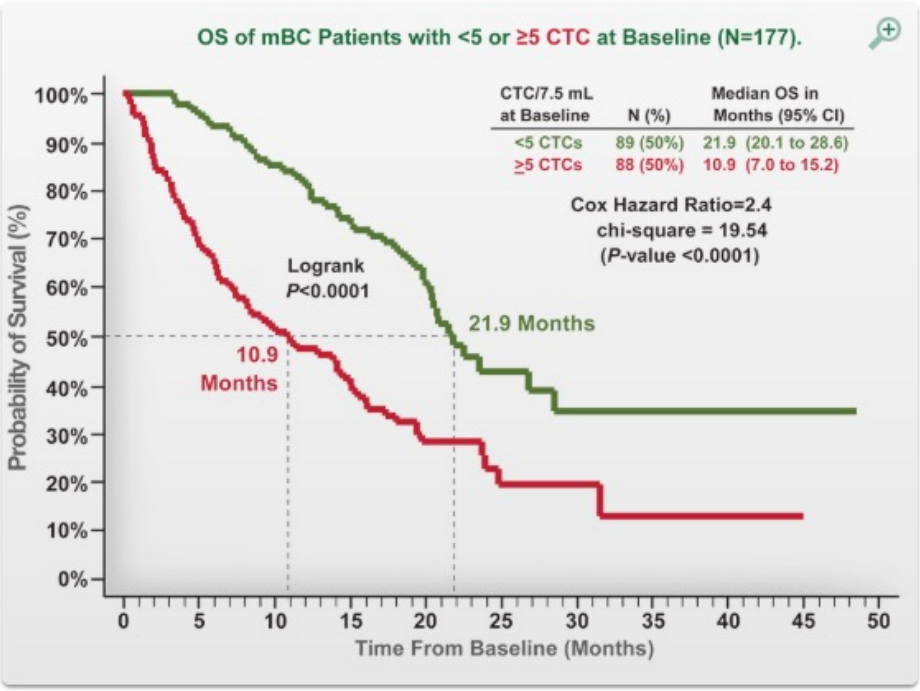
How CellSearch works?

- **Centrifuge to remove plasma**
- **Add Anti-Epcam FerrorFluidto capture cells (Immunomagnetic separation)**
- **Staining the cells with**
  - **DAPI for nucleus**
  - **CD45 for leukocytes**
  - **CK+ for epithelial cells**
- **Examine each one with fluorescence microscope**

Watch this video
https://www.youtube.com/watch?v=7Qi85f66oVA

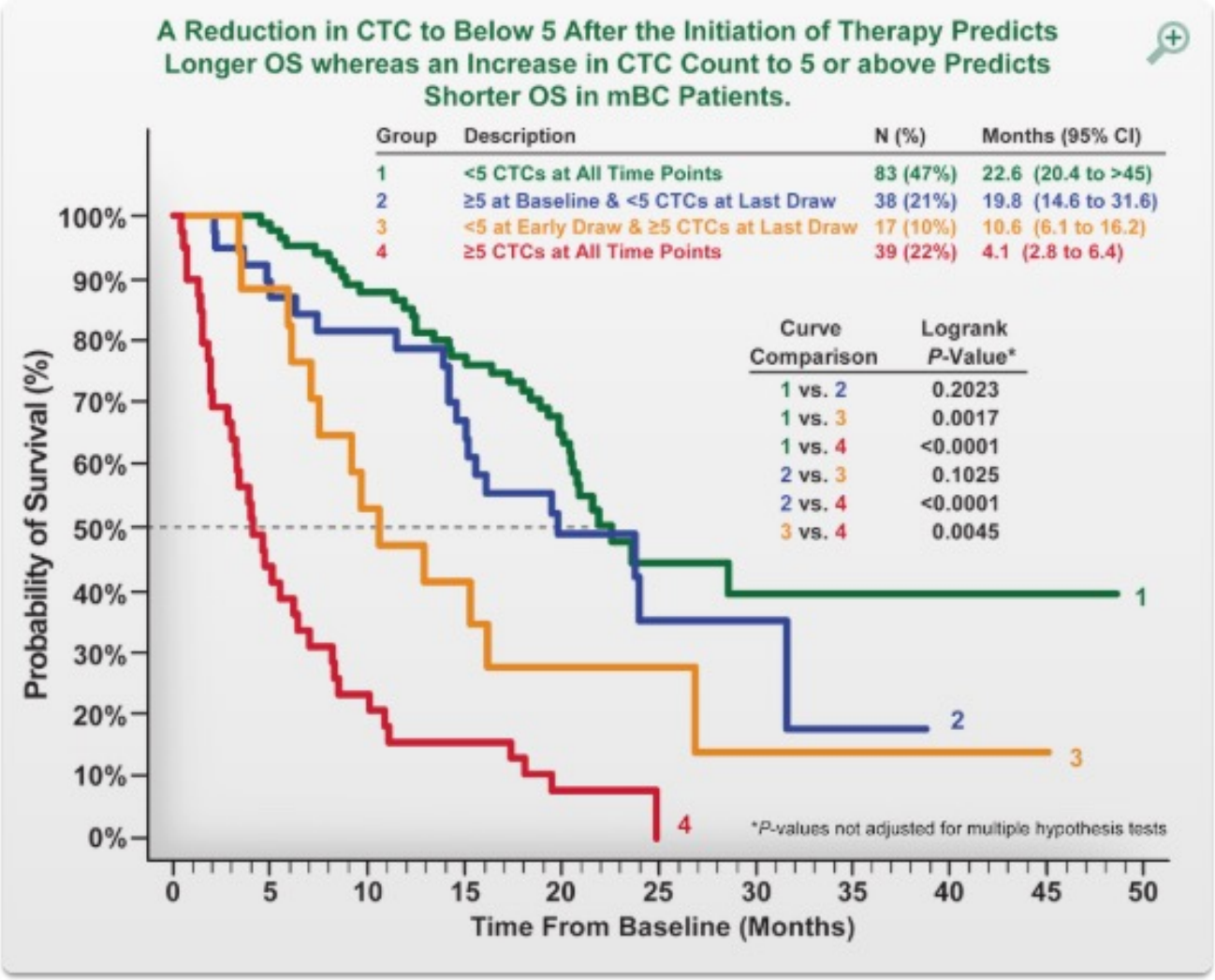


How does enumerating the number of CTCs help?



- Enrolled 177 mBC patients
- Kaplan-Meyer plot for overall survival for enrolled patients
  - Patients with <5 CTCs overall survival is 21.9 months
  - Patients with >5 CTCs have an overall survival of 10.9months
- Number of CTCs impact the overall survival

OS: overall survival
mBC: mestatic breast cancer


A Reduction in CTC to Below 5 After the Initiation of Therapy Predicts Longer OS whereas an Increase in CTC Count to 5 or above Predicts Shorter OS in mBC Patients.
Group Description N (%) Months (95% CI)
1 <5 CTCs at All Time Points 83 (47%) 22.6 (20.4 to >45)
2 ≥5 at Baseline & <5 CTCs at Last Draw 38 (21%) 19.8 (14.6 to 31.6)
3 <5 at Early Draw & ≥5 CTCs at Last Draw 17 (10%) 10.6 (6.1 to 16.2)
4 ≥5 CTCs at All Time Points 39 (22%) 4.1 (2.8 to 6.4)
Curve Comparison Logrank P-Value*
1 vs. 2 0.2023
1 vs. 3 0.0017
1 vs. 4 <0.0001
2 vs. 3 0.1025
2 vs. 4 <0.0001
3 vs. 4 0.0045
Probability of Survival (%)
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
0 5 10 15 20 25 30 35 40 45 50
Time From Baseline (Months)
1
2
3
4
*P-values not adjusted for multiple hypothesis tests

Kaplan-Meyer plot

- A Statistical method to estimate the probability of an event occurring at different points in time
- Frequently used for clinical trials to observe treatments effectiveness
- Survival curve is generated (non-parametric) by observing “events (i.e., deaths) at time intervals
- You need to follow up patients and know their survival time

In-Class activity

- Visit Xena browser, TCGA BRCA study
- Download patient survival data (under phenotype: Curated Survival data)
- Read the survival data using pandas—follow instructor's example
- You will successfully create the Kaplan Meier plot for TCGA BRCA using overall survival

Liquid Biopsy

- A **liquid biopsy** is a test done on a sample of blood (or sometimes other fluids like urine or cerebrospinal fluid) to detect diseases (for example cancer metastasis)
- CellSearch system is one kind of liquid biopsy technology
- There are other technology platforms developed

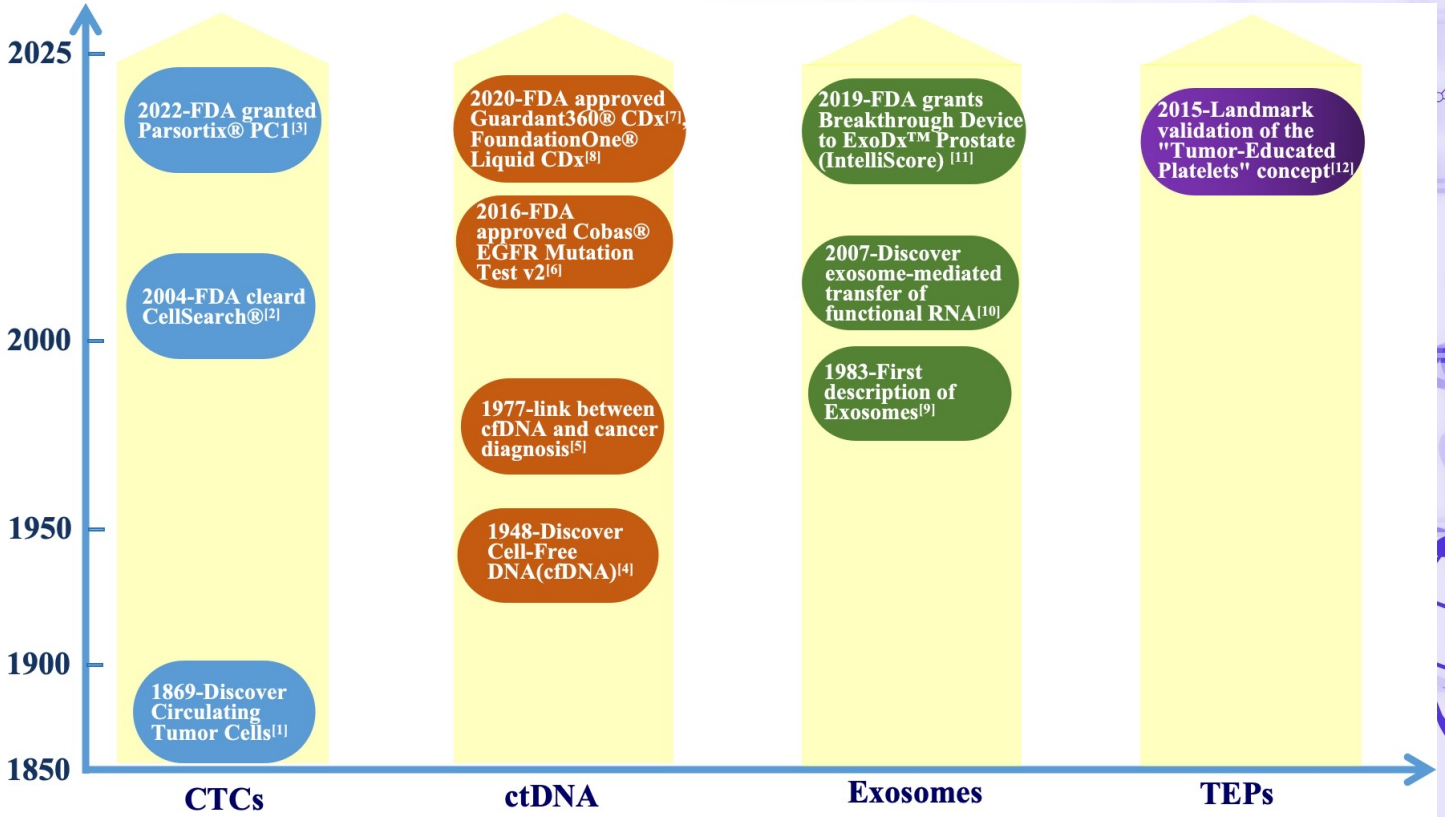
Why Liquid Biopsy in Metastasis?

- **Metastatic tumors are often hard to reach.**
- **Metastatic burden is dynamic**, and liquid biopsy allows repeated sampling over time.
- It provides a **non-invasive snapshot** of the tumor evolution, heterogeneity, and treatment resistance.

What Can Liquid Biopsy Detect?

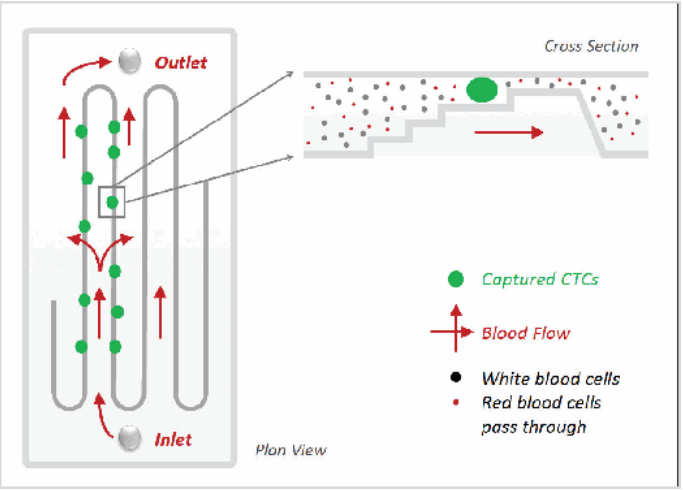
Biomarker Type	Description	Relevance to Metastasis
CTCs (Circulating Tumor Cells)	Tumor cells shed into bloodstream	Early signs of metastasis, potential to seed new tumors
ctDNA (circulating tumor DNA)	Fragments of tumor DNA	Mutation tracking, resistance evolution
cfRNA	RNA transcripts from tumors	Gene expression changes, fusion genes
Exosomes / EVs	Tumor-derived vesicles	Cargo (DNA, RNA, proteins) can mediate pre-metastatic niche formation
Tumor-Educated Platelets	Platelets altered by tumors	Emerging marker, reflects tumor presence

Some FDA-approved liquid biopsy Technology platforms



Parsortix versus CellSearch

- CellSearch capture CTCs via anti-EpCAM. This is a major limitation. Some cancer types do not have EpCAM expression
- Parsortix use microfluidic and cell deformability
  - Tumor cells are larger and less deformable than blood cells



Marketed by Angle PLC.
Publicly Traded: Listed on the **London Stock Exchange (AIM:AGL)**

ctDNA

- Circulating tumor DNA: fragments from DNA released by tumor into blood stream
- The company focused on developing detection of ctDNA **Guardant Health**
- **Guardant Health was co-founded by** Helmy Eltoukhy and AmirAli Talasaz during 2012–2013; both got Ph.D from Stanford

Helmy Eltoukhy

Alma mater	Stanford University BS, Electrical engineering MS, Electrical engineering PhD, Electrical engineering Stanford Genome Technology Center (Postdoc)

AmirAli Talasaz, PhD

Co-Chief Executive Officer

https://guardanthealth.com/

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Screen captured from the website: guardanthealth.com

Guardant Health Stock price

Market Summary > Guardant Health Inc

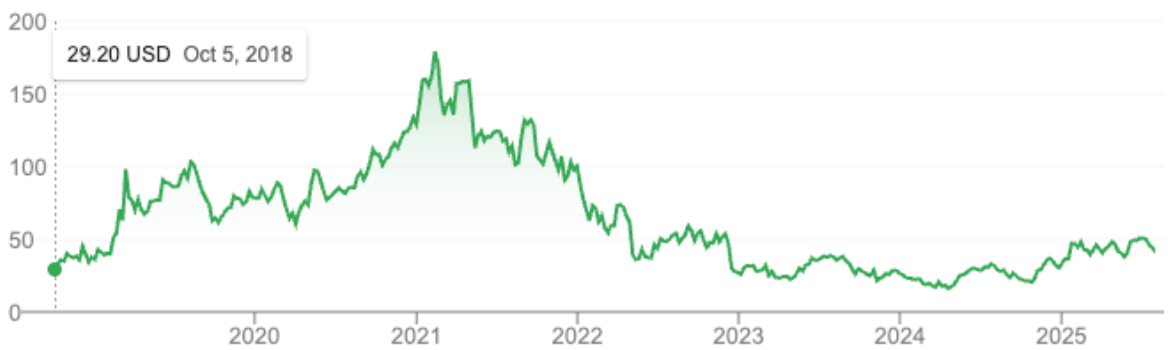
41.42 USD

+ Follow

+12.22 (41.85%) ↑ all time

Closed: Aug 1, 4:20 PM EDT • Disclaimer
After hours 41.42 0.00 (0.00%)

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max



Open	40.75	Mkt cap	5.17B	52-wk high	53.42
High	42.22	P/E ratio	-	52-wk low	20.14
Low	40.36	Div yield	-	Qtrly Div Amt	-

Become public traded in 2018

Future of Liquid Biopsy

- Transforming Cancer Detection and Monitoring
- From Late Detection to Early Detection
- Real-Time Monitoring of Tumor Evolution
- Non-Invasive and Broadly Accessible
- Integration with Big Data: Enabling Precision Medicine

Future of Liquid Biopsy and Big Data

“In the coming decade, your blood could tell you if you’re developing cancer—**before it even causes symptoms.** And data from millions of patients could help tailor treatment to you, not just the disease. That’s the power of combining **liquid biopsy with big data.**” –ChatGPT 4.0

Blogging

- Blog what you learned about cancer metastasis
  - Do some research to see the metastasis sites most frequently occurred for the cancer of your choice
- Discuss how one can detect cancer metastasis through liquid biopsy. Include some FDA-approved techniques that have been adopted for use in clinics and their prospects